Table 9.1.1: Schedule of Events-Screening and Infusion

Assessment	Prior to BMN 270 Infusion				BMN 270 Infusion Visit (Day 1)[o]
	Screening		Smart Rescreening[m] (Day -28 to Day -1)	Baseline (Day -7 to Day -1)[n]	
	Screening Day -42 to Day -29	Screening* (Day -28 to Day -1)			
Informed consent	X	X[p]			
Demographics (age, sex, race, ethnicity)		X			
Medical History		X			
Physical Examination[a]		X		X	X
Height and Weight		X			
Vital Signs		X	X	X	X
Assessment of Adverse Events and Concomitant Medications		X	X	X	X
Documentation of bleeding episodes and hemophilia therapy usage for previous 12 months (by either subject or clinical information)		X	X	X	
Distribution of subject diaries and training in their use[b]		X			
Electrocardiogram		X			
Liver Ultrasound (and FibroScan at discretion of Investigator)[c]		X			
hFVIII Assays[d]		X	X[q]	X	
hFVIII TAb		X		X	
Screen for Hepatitis B, Hepatitis C, HIV[e]		X			
SARS-CoV-2 screening (local or central)[f]		X	X		
Blood chemistry, hematology, and coagulation tests[g]		X	X	X	
Fasting lipid panel (blood triglycerides, total cholesterol, HDL cholesterol, and LDL cholesterol)					X
Fasting FibroTest		X			

Assessment	Prior to BMN 270 Infusion				BMN 270 Infusion Visit (Day 1)[o]
	Screening		Smart Rescreening[m] (Day -28 to Day -1)	Baseline (Day -7 to Day -1)[n]	
	Screening Day -42 to Day -29	Screening* (Day -28 to Day -1)			
Urine Tests[g]		X	X	X	
Liver Tests[g]		X	X	X	
AAV5 TAb Assay (CDx)[h]	X	X	X		X
Immunogenicity AAV5 TAb Assay				X	X
AAV5 TI Assay				X	X
IFNg ELISpot				X	
Plasma, PBMC, and RBC collection for exploratory biomarkers[i]				X	
Biomarker testing[i]		X			
Serum for exploratory biomarkers[j]		X			X
Exploratory CK18 and Grp78 assessment		X		X	
Haemo-QoL-A assessment				X	
EQ-5D-5L				X	
WPAI+CIQ:HS				X	
PROBE				X	
PCR of vector DNA in blood, saliva, urine, semen, and stools				X	X
Pharmacokinetics				X[r]	
BMN 270 Infusion					X
Complement Panel and Exploratory Cytokine Profiling[k]				X	X[k]
Hypersensitivity blood assessments[l]					X[l]

* Screening assessments should be performed within 28 days of BMN 270 infusion (and must be performed within 42 days prior to BMN 270 infusion).

[a] Complete physical examination should be done at Screening. Brief physical examination may be done at Baseline and at the BMN 270 Infusion Visit.

[b] Diaries should be distributed to subjects who have consented to participate in the study and who have been determined to meet all study eligibility criteria.

[c] All patients must have a liver ultrasound performed during the Screening period to screen for significant liver disease and hepatocellular carcinoma. A FibroScan may also be performed at the discretion of the Investigator.

[d] Includes baseline FVIII activity (chromogenic substrate FVIII assay and one-stage clotting FVIII assay), coagulation exploratory assay, chromogenic Nijmegen-Bethesda Assay for hFVIII inhibitor level, and hFVIII protein assay. Baseline activity should be assessed at trough (at least >72 hours after last dose of replacement FVIII therapy, or 5x the known half-life of the FVIII concentrates administered).

[e] Patients with documented negative results within the last 30 days do not need to be retested. Hepatitis B screening should include hepatitis B surface antigen (HBsAg), hepatitis B surface antibody (HBsAb), and hepatitis B core antibody (HBcAb).

[f] SARS-CoV-2 RT-PCR testing is required during Screening, and all subjects must have at least one negative test result prior to dosing. If the test is performed locally an additional 2nd test is recommended, but if performed negative results from the 2nd test must be received prior to dosing. A SARS-CoV-2 vaccine is recommended for subjects if indicated and if available. If a two-step SARS-CoV-2 vaccine is being used, sites should consider using the flexible re-screen option to allow subjects to receive both doses at least 14 days prior to treatment with BMN 270 (or at least 30 days prior to treatment with BMN 270 for any live-virus vaccines). It is preferable for SARS-CoV-2 vaccination to occur prior to BMN 270 infusion. Investigators should use clinical judgment, taking into consideration local factors, individual risk factors, and benefit/risk related to timing of vaccine administration.

[g] Refer to Table 9.7.8.2.1 for laboratory assessments to be included, and to Table 9.7.8.4.1 for liver tests. ABO blood typing assessment should be performed at Screening.

[h] Screening, Smart Re-screening, and Infusion Day samples will be tested using the CDx AAV5 total antibody (TAb) assay. During Screening, the CDx AAV5 TAb assay test may be done first, under a standalone informed consent form, before the main ICF for the study is signed and further screening procedures are performed. If performed during the early Screening period, the CDx AAV5 TAb assessment does not need to be repeated as part of general Screening. Sample collection on the day of the infusion visit must be performed before the BMN 270 infusion is given.

[i] Includes HLA genotyping and FVIII genotyping.

[j] Blood samples will be collected to evaluate biochemical, molecular, cellular, immunological, and genetic/genomic aspects relevant to hemophilia A, coagulation, and/or AAV gene transfer, and to develop assays used for these evaluations. Subjects may choose to opt out of the exploratory genetic/genomic research being done to study or try to discover genes that are not yet known to be associated with hemophilia A. While exploratory samples will be collected at the time points indicated above, testing of these samples will be performed only as deemed necessary by the Sponsor.

[k] Complement panel should include C3, C3a, C4, Bb, and sC5b-9 (refer to Table 9.7.8.2.1) and should be collected 2 hours after completion of the infusion. While exploratory samples for cytokine profiling will be collected at the time points indicated above, testing of these samples will be performed only as deemed necessary by the Sponsor.

[l] In case of a Grade 2 or higher hypersensitivity or adverse drug reaction, a safety assessment including physical examination and vital signs will be performed and additional blood samples will be collected within 1 hour, and 8-24 hours following the hypersensitivity reaction for assessment of complement (C3, C3a, C4, Bb, and sC5b-9) and tryptase. Additional samples will be collected at the 1 hour and 8-24 hour time points and, if possible, 1 week after the event for optional exploratory cytokine profiling to assess inflammatory biomarkers and plasma cytokine levels. Inpatient observation can be extended and additional blood samples can be collected if deemed necessary at the discretion of the Investigator and Medical Monitor.

[m] Smart rescreening should only be performed if a patient has been determined to be eligible for the study and is unable to complete the Baseline assessments and Infusion prior to the closing of the original Screening window. SARS-CoV-2 RT-PCR testing is required during Screening, and all subjects must have at least one negative test result prior to dosing. If the test is performed locally an additional 2nd test is recommended, but if performed negative results from the 2nd test must be received prior to dosing. Subjects who undergo smart rescreening

must complete the rescreening assessments and receive the infusion within 90 days of signing the original consent. Subjects who do not complete dosing within 90 days will be required to re-consent and undergo all screening procedures. Subjects may not undergo smart rescreening more than once.

[n] Should the screening visit occur within 30 days of infusion, physical examination, blood chemistry, LTs, hematology, and urine and coagulation tests do not need to be repeated at Baseline.

[o] With the exception of the collection of samples for polymerase chain reaction (PCR) vector DNA analysis and the collection of the complement panel/exploratory cytokine profiling sample, assessments on the day of infusion must be performed prior to the infusion. Subjects will fast for at least 8 hours prior to pre-infusion laboratory sampling on the day of the infusion visit. On the day of the BMN 270 Infusion, vital signs will be monitored prior to the infusion, during the infusion every 15 minutes (± 5 minutes), and following the infusion hourly (± 5 minutes) for at least 8 hours during the subject's stay in the clinic. Shedding samples for PCR of vector DNA analysis (blood, saliva, urine, semen, stool) should be collected between 2 and 24 hours after the infusion has been completed.

[p] If the subject underwent early AAV5 TAb testing and was consented using the full study ICF, the ICF does not need to be re-administered and re-signed as part of regular Screening. If the subject underwent early AAV5 TAb testing and was consented using the dedicated stand-alone ICF for that purpose, the full ICF will need to be signed if the subject proceeds to regular Screening.

[q] Only the hFVIII inhibitor level (Bethesda assay with Nijmegen modification) assay must be done at smart rescreening.

[r] Samples will be drawn immediately prior to recombinant FVIII concentrate infusion (between Day -2 and Day -7), 3 hours (+/- 30 minutes) post-FVIII infusion, and 24-52 hours post-FVIII infusion. For subjects receiving emicizumab, pharmacokinetics assessment is optional.

Table 9.1.2: Schedule of Events-Post-Infusion Follow-Up (Week 1-20)

Assessment	Follow-Up After BMN 270 Infusion – Weeks																			
	1	**2[f]**	**3[f]**	**4**	**5[f]**	**6[f]**	**7[f]**	**8**	**9[f]**	**10[f]**	**11[f]**	**12**	**13[f]**	**14[f]**	**15[f]**	**16**	**17[f]**	**18**	**19[f]**	**20**
Study Day[*]	8	15	22	29	36	43	50	57	64	71	78	85	92	99	106	113	120	127	134	141
Physical examination[a]				X				X				X				X		X		X
Weight				X				X				X				X				X
Assessment of Adverse Events and Concomitant Medications (including review of bleeding and FVIII use)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Vital Signs				X				X				X				X				X
Blood chemistry, hematology, and coagulation tests[b]				X												X				
Urine Tests[b]												X								
Liver Tests[b]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
hFVIII assays[c]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
hFVIII protein assay				X				X				X				X				X
PCR of vector DNA in blood, saliva, urine, semen, and stools[d]	X			X				X				X				X				X
Immunogenicity AAV5 TAb Assay				X				X								X				
AAV5 TI Assay				X				X								X				
hFVIII TAb				X				X				X				X				X
IFNg ELISpot		X				X				X				X				X		
Plasma, PBMC, and RBC collection for exploratory biomarkers[e]	X			X				X				X						X		
Complement Panel and Exploratory Cytokine Profiling[b]	X	X		X				X				X				X				X

Assessment	Follow-Up After BMN 270 Infusion – Weeks																			
	1	**2[f]**	**3[f]**	**4**	**5[f]**	**6[f]**	**7[f]**	**8**	**9[f]**	**10[f]**	**11[f]**	**12**	**13[f]**	**14[f]**	**15[f]**	**16**	**17[f]**	**18**	**19[f]**	**20**
Study Day[*]	8	15	22	29	36	43	50	57	64	71	78	85	92	99	106	113	120	127	134	141
Serum for exploratory biomarkers[e]	X	X		X		X		X		X		X				X				X
Exploratory CK18 assessment	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Haemo-QOL-A assessment				X								X								
EQ-5D-5L				X								X								
WPAI+CIQ:HS				X								X								
PROBE				X								X								
Testing for reactivation of hepatitis B and hepatitis C						X[g]														X[g]

[*] Visit windows are ± 48 hours.

[a] Brief physical examination should be done at scheduled visits. Additional physical exams may be done at the discretion of the PI.

[b] Refer to Table 9.7.8.2.1 for laboratory assessments to be included and for complement panel tests, and to Table 9.7.8.4.1 for liver tests (LTs). LT assessment is weekly, but may be checked more frequently when ALT values are > ULN or ≥ 1.5x baseline value or based upon discussion between the Medical Monitor and the Investigator. Subjects with ALT > ULN or ≥ 1.5x baseline value during the study and either an alternative etiology considered or for further evaluation of the lab abnormality may undergo additional evaluations (eg, imaging studies including MRI or ultrasound, additional blood tests, and/or liver biopsy) at the discretion of the Investigator. Additional testing of FVIII and LTs during any study week may be performed if: the ALT has increased to ≥ 1.5x baseline value or >ULN or after a discussion between the Medical Monitor and the Investigator based on a review of subject data. If ALT values are stable over the course of several weeks, assessment of FVIII activity and ALT levels may be adjusted based on discussion between the Medical Monitor and the Investigator. In case of a safety concern, additional unscheduled laboratory tests may be performed locally (at the Investigator's discretion) in order to facilitate timely and appropriate clinical management decisions; where possible, a matched sample for testing at the central laboratory should be collected (using the Unscheduled Visit lab kit) at the same time as the local unscheduled sample. While exploratory samples for cytokine profiling will be collected at the time points indicated above, testing of these samples will be performed only as deemed necessary by the Sponsor.

[c] Includes FVIII activity level (chromogenic substrate FVIII assay and one-stage clotting FVIII assay), chromogenic Nijmegen-Bethesda Assay for hFVIII inhibitor level, and coagulation exploratory assay. Chromogenic Nijmegen-Bethesda Assay for hFVIII inhibitor level will be tested as deemed necessary by the Sponsor. If a subject has used FVIII within 72 hours of a measurement day, all efforts should be made to obtain FVIII activity measurements when a 72-hour interval without FVIII use is achieved; the 72-hour wash-out period is only intended for subjects who have achieved FVIII ≥ 5 IU/dL at 16 weeks after BMN 270 infusion and who have not resumed FVIII replacement therapy. If FVIII levels are elevated or stable over the course of several weeks, assessment of FVIII activity may be adjusted based on discussion between the Medical Monitor and the Investigator. A D-dimer may be considered under circumstances in which FVIII activity levels are above the normal physiological range and/or if there is a concern for a venous thromboembolism.

[d] Collection for each matrix to occur until at least 3 consecutive results below the limit of detection are obtained; longer collection and testing may be performed based on batch testing schedules, result turnaround times, or discussions between Medical Monitor and Investigator. Collection and testing of semen samples must continue at least through Week 12, even if 3 consecutive results below the limit of detection in that compartment have already been recorded.

[e] Blood samples will be collected to evaluate biochemical, molecular, cellular, and genetic/genomic aspects relevant to hemophilia A, coagulation, and/or AAV gene transfer, and to develop assays used for these evaluations. Subjects may choose to opt out of the exploratory genetic/genomic research being done to study or try to discover genes that are not yet known to be associated with hemophilia A. While exploratory samples will be collected at the time points indicated above, testing of these samples (including those for any other exploratory assessments) will be performed only as deemed necessary by the Sponsor.

[f] Visits between scheduled clinic visits may be performed by a mobile nursing (MN) professional at the subject's home or another suitable location (if the subject has given written informed consent to participate in MN visits), or at the site or approved lab facility as a shortened lab draw-only visit. The MN service or site will collect blood samples at these visits. For site lab draw-only visits, sites will contact the subject via phone call or e-mail to collect information regarding adverse events, changes to concomitant medications, bleeding episodes, and FVIII use. For MN visits, the service will collect this information. Unscheduled visits may also be conducted by MN as appropriate.

[g] Testing for reactivation of hepatitis B and hepatitis C only for subjects with a past medical history of hepatitis B or hepatitis C prior to study entry.

Table 9.1.3: Schedule of Events – Post-Infusion Follow-Up (Weeks 21-52)

Assessment	Follow-Up After BMN 270 Infusion-Weeks															
	21[f]	**22**	**23[f]**	**24**	**25[f]**	**26**	**28**	**30[f]**	**32**	**34[f]**	**36**	**40**	**44**	**48**	**50[f]**	**52**
Study Day[*]	148	155	162	169	176	183	197	211	225	239	253	281	309	337	351	365
Physical examination[a]		X		X		X	X		X		X	X	X	X		X
Weight						X			X		X		X			X
Assessment of Adverse Events and Concomitant Medications (including review of subject diary for bleeding and FVIII use)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Vital Signs				X		X			X		X	X	X	X		X
Blood chemistry, hematology, and coagulation tests[b]						X			X		X		X			X
Urine Tests[b]						X					X					X
Liver Tests[b]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
hFVIII assays[c]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
hFVIII protein assay				X		X			X		X	X	X	X		X
PCR of vector DNA in blood, saliva, urine, semen, and stools[d]						X			X		X		X			X
Immunogenicity AAV5 TAb Assay						X					X					X
AAV5 TI Assay						X					X					X
hFVIII TAb				X		X					X					X
IFNg ELISpot		X		X		X					X					X
Plasma, PBMC, and RBC collection for exploratory biomarkers[e]		X				X	X				X		X			X
Complement Panel and Exploratory Cytokine Profiling[b]				X		X					X					X
Serum for exploratory biomarkers[e]				X		X					X					X
Exploratory CK18 assessment	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Assessment	Follow-Up After BMN 270 Infusion-Weeks															
	21[f]	22	23[f]	24	25[f]	26	28	30[f]	32	34[f]	36	40	44	48	50[f]	52
Study Day[*]	148	155	162	169	176	183	197	211	225	239	253	281	309	337	351	365
Haemo-QOL-A assessment						X										X
EQ-5D-5L						X										X
WPAI+CIQ:HS						X										X
PROBE						X										X
Testing for reactivation of hepatitis B and hepatitis C									X[g]							
Optional liver biopsy[h]						X										X
Liver ultrasound[i]																X

[*] Visit windows are ± 48 hours.

[a] Brief physical examination should be done at all visits where a physical examination is indicated except Week 26 and Week 52, where a complete physical examination should be performed Additional physical exams may be done at the discretion of the PI.

[b] Refer to Table 9.7.8.2.1 for laboratory assessments to be included and for complement panel tests, and to Table 9.7.8.4.1 for liver tests (LTs). LT assessment may be checked more frequently when ALT values are > ULN or ≥ 1.5x baseline value or based upon discussion between the Medical Monitor and the Investigator. Subjects with > ULN or ≥ 1.5x baseline value during the study and either an alternative etiology considered or for further evaluation of the lab abnormality may undergo additional evaluations (eg, imaging studies including MRI or ultrasound, additional blood tests, and/or liver biopsy) at the discretion of the Investigator. Additional testing of FVIII and LTs during any study week may be performed if: the ALT has increased to ≥ 1.5x baseline value or >ULN or after a discussion between the Medical Monitor and the Investigator based on a review of subject data. If FVIII levels and/or ALT values are stable over the course of several weeks, assessment of FVIII activity and ALT levels may be adjusted based on discussion between the Medical Monitor and the Investigator. In case of a safety concern, additional unscheduled laboratory tests may be performed locally (at the Investigator's discretion) in order to facilitate timely and appropriate clinical management decisions; where possible, a matched sample for testing at the central laboratory should be collected (using the Unscheduled Visit lab kit) at the same time as the local unscheduled sample. While exploratory samples for cytokine profiling will be collected at the time points indicated above, testing of these samples will be performed only as deemed necessary by the Sponsor.

[c] Includes FVIII activity level (chromogenic substrate FVIII assay and one-stage clotting FVIII assay), chromogenic Nijmegen-Bethesda Assay for hFVIII inhibitor level, and coagulation exploratory assay. Chromogenic Nijmegen-Bethesda Assay for hFVIII inhibitor level will be tested as deemed necessary by the Sponsor. If a subject has used FVIII within 72 hours of a measurement day, all efforts should be made to obtain FVIII activity measurements when a 72-hour interval without FVIII use is achieved; the 72-hour wash-out period is only intended for subjects who have achieved FVIII ≥ 5 IU/dL at 16 weeks after BMN 270 infusion and who have not resumed FVIII replacement therapy. If FVIII levels are elevated or stable over the course of several weeks, assessment of FVIII activity may be adjusted based on discussion between the Medical Monitor and the Investigator. A D-dimer may be considered under circumstances in which FVIII activity levels are above the normal physiological range and/or if there is a concern for a venous thromboembolism.

[d] Collection for each matrix to occur until at least 3 consecutive results below the limit of detection are obtained; longer collection and testing may be performed based on batch testing schedules, result turnaround times, or discussions between Medical Monitor and Investigator.

[e] Blood samples will be collected to evaluate biochemical, molecular, cellular, and genetic/genomic aspects relevant to hemophilia A, coagulation, and/or AAV gene transfer, and to develop assays used for these evaluations. Subjects may choose to opt out of the exploratory genetic/genomic research being done to study or try to discover genes that are not yet known to be associated with hemophilia A. While exploratory samples will be collected at the time points indicated above, testing of these samples (including those for any other exploratory assessments) will be performed only as deemed necessary by the Sponsor.

[f] Visits between scheduled clinic visits may be performed by a mobile nursing (MN) professional at the subject's home or another suitable location (if the subject has given written informed consent to participate in MN visits), or at the site or approved lab facility as a shortened lab draw-only visit. The MN service or site will collect blood samples at these visits. For site lab draw-only visits, sites will contact the subject via phone call or e-mail to collect information regarding adverse events, changes to concomitant medications, bleeding episodes, and FVIII use. For MN visits, the service will collect this information. Unscheduled visits may also be conducted by MN as appropriate.

[g] Testing for reactivation of hepatitis B and hepatitis C only for subjects with a past medical history of hepatitis B or hepatitis C prior to study entry.

[h] Optional liver biopsy should be obtained within two weeks of Week 26. Additional follow-up liver biopsy will be obtained within two weeks of Week 52. Subjects should fast for at least 8 hours prior to liver ultrasound and optional liver biopsies.

[i] Additional liver ultrasounds may be performed prior to Week 52 at the discretion of the Investigator.

Table 9.1.4: Schedule of Events-Post-Infusion Follow-Up (Year 2-Year 5)

Assessment	Years 2-5*			End of Year Visit				ETV
	Years 2-5	Year 2	Year 3-5	Year 2	Year 3	Year 4	Year 5	
Study Week[*]	Q12W	Q4W[g]	Q6W[g]	W104	W156	W208	W260	
Physical examination[b]	X			X				X
Weight[b]	X[a]			X[a]				X
Assessment of Adverse Events and Concomitant Medications (including review of subject diary for bleeding and hemophilia therapy use)	X	X	X	X				X
Vital Signs	X			X				X
Blood chemistry, hematology, and coagulation tests[c]	X[a]			X[c]				X
Urine Tests[c]	X[a]			X[c]				X
Liver Tests[c]	X	X	X	X				X
hFVIII assays[d]	X	X	X	X				X
hFVIII protein assay	X			X				X
PCR of vector DNA in blood, saliva, urine, and stools	(X)[e]			(X)[e]				(X)[e]
PCR of vector DNA in semen	(X)[e]	(X)[e]	(X)[e]	(X)[e]				(X)[e]
Immunogenicity AAV5 TAb Assay				X				X
AAV5 TI Assay				X[j]				X[j]
hFVIII TAb	X			X				X
IFNg ELISpot[a]	X[a]			X				X
Plasma, PBMC, and RBC collection for exploratory biomarkers[f]	X			X				X
Serum for exploratory biomarkers[f]	X			X				X
Exploratory CK18 assessment	X	X	X	X				X
Haemo-QoL-A assessment	X[a]			X[a]				X
EQ-5D-5L	X[a]			X[a]				X

Assessment	Years 2-5*			End of Year Visit				ETV
	Years 2-5	Year 2	Year 3-5	Year 2	Year 3	Year 4	Year 5	
Study Week*	Q12W	Q4W[g]	Q6W[g]	W104	W156	W208	W260	
WPAI+CIQ:HS	X[a]			X[a]				X
PROBE	X[a]			X[a]				X
Optional liver biopsy[h]				X				
Liver ultrasound[i]				X				

* Visit windows are ± 2 weeks for visits in Years 2-5.

[a] These assessments need to be performed only at every other Q12W and every End of Year visit (ie, Weeks 76, 104, 128, 156, 180, 208, 232, and 260).

[b] Complete physical examination should be performed at the End of Year visits (genitourinary examination may be deferred); brief physical exam may be performed at other study visits. Weight should be recorded at the second Q12W visit each year and at every End of Year visit during Years 2-5.

[c] Refer to Table 9.7.8.2.1 for laboratory assessments to be included, and to Table 9.7.8.4.1 for liver tests. LT assessment may be checked more frequently when ALT values are > ULN or ≥ 1.5x baseline value or based upon discussion between the Medical Monitor and the Investigator. Subjects with ALT > ULN or ≥ 1.5x baseline value during the study and either an alternative etiology considered or for further evaluation of the lab abnormality may undergo additional evaluations (eg, imaging studies including MRI or ultrasound, additional blood tests, and/or liver biopsy) at the discretion of the Investigator. Additional testing of FVIII and LTs during any study week may be performed if: the ALT has increased to ≥ 1.5x baseline value or >ULN or after a discussion between the Medical Monitor and the Investigator based on a review of subject data. If FVIII levels and/or ALT values are stable over the course of several weeks, assessment of FVIII activity and ALT levels may be adjusted based on discussion between the Medical Monitor and the Investigator. During Years 2-5 of the Post-Infusion Follow-Up period, urine tests and blood, chemistry, and coagulation tests should be performed at the second Q12W visit each year and at every End of Year visit. In case of a safety concern, additional unscheduled laboratory tests may be performed locally (at the Investigator's discretion) in order to facilitate timely and appropriate clinical management decisions; where possible, a matched sample for testing at the central laboratory should be collected (using the Unscheduled Visit lab kit) at the same time as the local unscheduled sample

[d] Includes FVIII activity level (chromogenic substrate FVIII assay and one-stage clotting FVIII assay), chromogenic Nijmegen-Bethesda Assay for hFVIII inhibitor level, coagulation exploratory assay. Chromogenic Nijmegen-Bethesda Assay for hFVIII inhibitor level will be tested as deemed necessary by the Sponsor. If a subject has used FVIII within 72 hours of a measurement day, all efforts should be made to obtain FVIII activity measurements when a 72-hour interval without FVIII use is achieved; the 72-hour wash-out period is only intended for subjects who have achieved FVIII ≥ 5 IU/dL at 16 weeks after BMN 270 infusion and who have not resumed FVIII replacement therapy. If FVIII levels are elevated or stable over the course of several weeks, assessment of FVIII activity may be adjusted based on discussion between the Medical Monitor and the Investigator. A D-dimer may be considered under circumstances in which FVIII activity levels are above the normal physiological range and/or if there is a concern for a venous thromboembolism. If a subject tests positive in the Bethesda assay (with Nijmegen modification) during Years 3-5, a fresh sample should be collected within 4 weeks of the initial visit that had a positive result.

[e] Sample testing during Years 2-5 is not required if at least 3 consecutive samples were below the limit of detection during Year 1; additional collection and testing may be performed based on batch testing schedules, result turnaround times, or discussions between Medical Monitor and Investigator. Subjects who have not had 3 consecutive semen samples below the limit of detection by Week 52 should continue to have PCR testing of semen every 4 weeks (during Year 2) or every 6 weeks (during Years 3-5) until 3 consecutive samples below the limit of detection are documented (or upon consultation between the Investigator and Medical Monitor).

[f] Blood samples will be collected to evaluate biochemical, molecular, cellular, immunological, and genetic/genomic aspects relevant to hemophilia A, coagulation, and/or AAV gene transfer, and to develop assays used for these evaluations. Subjects may choose to opt out of the exploratory genetic/genomic research being done to study or try to discover genes that are not yet known to be associated with hemophilia A. While exploratory samples will be collected at the time points indicated above, testing of these samples will be performed only as deemed necessary by the Sponsor.

[g] Subjects who meet the definition of treatment failure to BMN 270 therapy after Week 52 may omit the Q4W and Q6W visits during Years 2-5, and must attend only the Q12W and End of Year visits. Subjects who are not attending the Q4W/Q6W visits during Years 2-5 may receive a scheduled monthly follow-up telephone call from the site to ask about adverse events, concomitant medications, and bleeding events/FVIII replacement usage. Such subjects following the abbreviated schedule who have not yet cleared vector shedding in semen must still provide samples Q4W (during Year 2) or Q6W (during Years 3-5) until vector shedding has been cleared, either by reporting to the site to provide samples or by providing those samples to a MN professional. Unscheduled visits may also be conducted by MN as appropriate.

[h] An optional liver biopsy may be performed at any time between Years 2-5 of the study. The optional biopsy would be triggered by a FVIII activity decline by > 50% from steady-state, over 2 consecutive measurements, or by a sustained ALT rise > ULN. If neither trigger is observed, the optional biopsy should be performed at the end of Year 5.

[i] Additional liver ultrasounds may be performed at interim timepoints (ie, between the End of Year visits) at the discretion of the Investigator.

[j] AAV5 TI Assay should be performed only at the end of the study (either the Year 5 End of Year Visit, or at the ETV for subjects who withdraw prior to the end of Year 5).

Table 9.1.5: Schedule of Events-Optional Liver Biopsy

	Within 28 Days Before Biopsy Day	Within 7 Days Before Biopsy Day	Biopsy Day (BD)
Informed Consent for Liver Biopsy Procedure	X		
Liver Ultrasound[a]	X		
Physical examination	X		X
Hematology, Coagulation, Chemistry Assessments[b]	X		X
Liver Tests[b]	X		X
FibroScan	X		
FVIII Activity Level Assessment (central and local)		X	X[*]
Exploratory CK18 and Grp78 assessment		X	X[*]
Pre-Biopsy Consultation[c]		X	
Liver Biopsy[d]			X
PBMC Collection (whole blood draw)			X[e]

[*] If the Day -7 and biopsy day visits occur on the same day, these tests do not need to be duplicated.

[a] Liver ultrasound must be performed within 28 days prior to the scheduled biopsy. Subjects should fast for at least 8 hours prior to liver ultrasound.

[b] Refer to Table 9.7.8.2.1 for laboratory assessments to be included, and to Table 9.7.8.4.1 for liver tests.

[c] Subjects will undergo a pre-biopsy consultation with the Investigator (treating hematologist) and hepatologist and/or radiologist.

[d] Subjects should fast for at least 8 hours prior to optional liver biopsy. Biopsy will be a percutaneous or transjugular biopsy under ultrasound guidance, performed according to the standard procedure of the institution. If only a small amount of tissue (< 2 cm) is obtained at the time of the biopsy, the subject may be asked to consent for a second pass. In this case, the original < 2 cm sample should still be retained and handled according to the instructions for handling biopsy specimens in the Laboratory Manual. Following completion of the biopsy, the subject should remain in the hospital under observation for at least 4-6 hours. Overnight post-procedure observation may be done at the Investigator's discretion.

[e] Blood draw for PBMC collection should be performed on the biopsy day or ± 1 week from the biopsy day.

Table 9.1.6: Suggested Schedule of Events-Prophylactic Corticosteroids

	Corticosteroid Treatment Period[b]																			Post-Corticosteroid Period[c]				
		Week																		Week				
Assessment	**D1**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**32**
Prophylactic corticosteroid dose (mg/day)	40 mg	40 mg	40 mg	40 mg	40 mg	40 mg	40 mg	40 mg	40 mg	35 mg	35 mg	30 mg	30 mg	25 mg	25 mg	20 mg	20 mg	15 mg	10 mg	5 mg					
FVIII activity testing	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Liver tests	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Hepatitis B testing[d]							X														X				X
HCV Viral Load[d]							X														X				X

[a] This table provides an example of a prophylactic corticosteroid course. Clinical judgment, weighing the potential risks and benefits of corticosteroid treatment, should always be exercised when considering adjustment of corticosteroid doses, and discussions between the Investigator and Medical Monitor are advised for any questions or concerns. Dosages are for prednisone or an equivalent dose of another corticosteroid. FVIII and liver tests are also reflected in the Schedule of Events for the study.

[b] Following initiation or completion of corticosteroid regimen, if a recurrence of ALT values > ULN or ≥ 1.5x baseline value is reported, corticosteroid management decisions will be based on discussions between the Investigator and Medical Monitor. Modification of the corticosteroid regimen may take into consideration timing of ALT elevation (prior to or after Week 52), as well as possible confounders for the ALT elevation and adverse events related to corticosteroid dosing. Guidance for tapering oral corticosteroid dosing can be found in Section 9.4.8.2, although a discussion between the PI and Medical Monitor should take place prior to tapering the corticosteroid dose.

[c] After discontinuation of oral corticosteroids, weekly labs for ALT levels and FVIII activity will be measured once a week for 4 weeks to ensure stability in values.

[d] Should only be performed in subjects with a history of hepatitis B or hepatitis C prior to study entry. Tests are also reflected in the Schedule of Events for the study.